

```

From Coq Require Import Arith Lia Reals ZArith Ring Lra.
From Coq Require Import Arith.PeanoNat.

```

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(* ===== *)
(* Прочтение рукописи Деденко через совместимость симметрии *)
(* коэффициентов, версия 15. *)
(* Параметры m,r пробегают вещественные числа; целочисленные *)
(* и модульные вопросы рассматриваются отдельно. Нечётное *)
(* биномиальное ядро имеет две вещественные нормализации: *)
(* core = n*1^n и core = 2^(n-1)*q^n. *)
(* Их сравнение даёт 1^n = q^n * 2^(n-1)/n. *)
(* Препятствие совместимости имеет вид: *)
(* 1^n = q^n тогда и только тогда, когда n = 2^(n-1), *)
(* а из n = 2^(n-1) следует n in {1,2}. *)
(* ===== *)

```

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(* ----- Вспомогательные факты о делимости биномиальных выражений ----- *)
Local Open Scope Z_scope.

```

```

Lemma Z_sub_split (a b c : Z) : a - b = (a - c) + (c - b).
Proof. ring. Qed.

```

```

Lemma Zpow_diff_divides (x y : Z) (n : nat) :
  Z.divide (x - y) (Z.pow x (Z.of_nat n) - Z.pow y (Z.of_nat n)).
Proof.

```

```

  induction n as [|n IH].
  - simpl. exists 0%Z; ring.
  - replace (Z.of_nat (S n)) with (Z.of_nat n + 1) by lia.
    rewrite !Z.pow_add_r by lia.
    rewrite !Z.pow_1_r.
    set (ux := Z.pow x (Z.of_nat n)) in *.
    set (uy := Z.pow y (Z.of_nat n)) in *.
    specialize (IH) as [r Hr].
    rewrite Z.mul_comm with (n := ux) (m := x).
    rewrite Z.mul_comm with (n := uy) (m := y).
    rewrite (Z_sub_split (x * ux) (y * uy) (x * uy)).
    assert (Hx : x * ux - x * uy = x * (ux - uy)) by ring.
    assert (Hy : x * uy - y * uy = (x - y) * uy) by ring.
    rewrite Hx, Hy.
    rewrite Hr.
    exists (x * r + uy).
    ring.

```

Qed.

```

Lemma odd_n_diff_pow_even
  (a b : Z) (n : nat) :
  Nat.odd n = true ->
  Z.divide 2 (Z.pow (a + b) (Z.of_nat n) - Z.pow (a - b) (Z.of_nat n)).
Proof.

```

```

  intros Hodd.
  apply Nat.odd_spec in Hodd.
  destruct Hodd as [k Hk].
  subst n.
  replace (Z.of_nat (2 * k + 1)) with (Z.of_nat (S (2 * k))) by lia.
  specialize (Zpow_diff_divides (a + b) (a - b) (S (2 * k))) as Hdiv.
  replace ((a + b) - (a - b)) with (2 * b) in Hdiv by ring.
  destruct Hdiv as [r Hr].

```

```

exists (b * r).
rewrite Hr.
ring.
Qed.

```

```

Close Scope Z_scope.

```

```

(* ----- Тождества с вещественными параметрами (m,p in R) ----- *)
Local Open Scope R_scope.

```

```

Lemma sum_diff_from_parameters_R
  (n : nat) (m p : R) :
  let z := pow m n + pow p n in
  let x := pow m n - pow p n in
  z + x = 2 * pow m n /\
  z - x = 2 * pow p n.

```

```

Proof.
  intros z x; unfold z, x; split; ring.
Qed.

```

```

Close Scope R_scope.

```

```

(* ----- Специализация на целочисленные параметры (m,p in Z) ----- *)
Local Open Scope Z_scope.

```

```

Lemma sum_diff_from_parameters_Z
  (n : nat) (m p : Z) :
  let z := m ^ Z.of_nat n + p ^ Z.of_nat n in
  let x := m ^ Z.of_nat n - p ^ Z.of_nat n in
  z + x = 2 * m ^ Z.of_nat n /\
  z - x = 2 * p ^ Z.of_nat n.

```

```

Proof.
  intros z x; unfold z, x; split; ring.
Qed.

```

```

Corollary parity_condition_Z
  (n : nat) (m p : Z) :
  let z := m ^ Z.of_nat n + p ^ Z.of_nat n in
  let x := m ^ Z.of_nat n - p ^ Z.of_nat n in
  Z.even (z + x) = true /\
  Z.even (z - x) = true.

```

```

Proof.
  intros z x.
  destruct (sum_diff_from_parameters_Z n m p) as [Hxz Hxz'].
  split.
  - replace (z + x) with (2 * m ^ Z.of_nat n) by exact Hxz.
    rewrite Z.even_mul; simpl; reflexivity.
  - replace (z - x) with (2 * p ^ Z.of_nat n) by exact Hxz'.
    rewrite Z.even_mul; simpl; reflexivity.
Qed.

```

```

Lemma no_parameters_if_parity_violation (n : nat) (z x : Z) :
  Z.even (z + x) = false /\ Z.even (z - x) = false ->
  ~ (exists m p : Z,
    z = m ^ Z.of_nat n + p ^ Z.of_nat n /\
    x = m ^ Z.of_nat n - p ^ Z.of_nat n).

```

```

Proof.

```

```

intros Hpar [m [p [Hz Hx]]].
destruct (sum_diff_from_parameters_Z n m p) as [Hsum Hdiff].
destruct Hpar as [H1|H2].
- rewrite Hz, Hx, Hsum in H1.
  rewrite Z.even_mul in H1; simpl in H1. discriminate.
- rewrite Hz, Hx, Hdiff in H2.
  rewrite Z.even_mul in H2; simpl in H2. discriminate.
Qed.

```

```

Lemma no_parameters_if_odd (n : nat) (z x : Z) :
  Z.odd (z + x) = true /\ Z.odd (z - x) = true ->
  ~ (exists m p : Z,
    z = m ^ Z.of_nat n + p ^ Z.of_nat n /\
    x = m ^ Z.of_nat n - p ^ Z.of_nat n).

```

```

Proof.
  intros Hodd [m [p [Hz Hx]]].
  destruct (sum_diff_from_parameters_Z n m p) as [Hsum Hdiff].
  destruct Hodd as [H1|H2].
  - rewrite Hz, Hx, Hsum in H1.
    rewrite Z.odd_mul in H1; simpl in H1. discriminate.
  - rewrite Hz, Hx, Hdiff in H2.
    rewrite Z.odd_mul in H2; simpl in H2. discriminate.
Qed.

```

```

Lemma no_parameters_for_example :
  ~ (exists m p : Z,
    2%Z = m ^ Z.of_nat 3 + p ^ Z.of_nat 3 /\
    1%Z = m ^ Z.of_nat 3 - p ^ Z.of_nat 3).

```

```

Proof.
  apply (no_parameters_if_parity_violation 3 2 1).
  now left.
Qed.

```

Close Scope Z_scope.

(* ----- Проверки шагов реконструкции рукописи ----- *)
Local Open Scope nat_scope.

```

Lemma equation_rearrange_nat (n x y z : nat) :
  (x ^ n + y ^ n = z ^ n)%nat -> (z ^ n - x ^ n = y ^ n)%nat.

```

```

Proof.
  intro H.
  rewrite <- H.
  lia.
Qed.

```

```

Lemma parameter_equations_raise_nat (n m p z x : nat) :
  z = (m ^ n + p ^ n)%nat ->
  x = (m ^ n - p ^ n)%nat ->
  (z ^ n = (m ^ n + p ^ n) ^ n /\
  x ^ n = (m ^ n - p ^ n) ^ n)%nat.

```

```

Proof.
  intros Hz Hx.
  subst z x.
  split; reflexivity.
Qed.

```

```

Lemma parameterized_difference_substitution_nat (n m p y : nat) :
  ((m ^ n + p ^ n) ^ n - (m ^ n - p ^ n) ^ n)%nat = y ^ n)%nat ->
  let z := (m ^ n + p ^ n)%nat in
  let x := (m ^ n - p ^ n)%nat in
  (z ^ n - x ^ n = y ^ n)%nat.

```

Proof.

```
intro H.
```

```
exact H.
```

Qed.

```

Lemma pythagorean_parameter_example :

```

```

  let m := 3 in let p := 2 in
  let z := (m ^ 2 + p ^ 2)%nat in
  let x := (m ^ 2 - p ^ 2)%nat in
  let y := (2 * m * p)%nat in
  (z ^ 2 = x ^ 2 + y ^ 2)%nat.

```

Proof.

```
simpl.
```

```
reflexivity.
```

Qed.

```
(* ----- Совместимость симметрии коэффициентов, версия 15 ----- *)
```

```
Local Open Scope R_scope.
```

```

Definition real_core_factorization (n : nat) (core l : R) : Prop :=
  (0 < n)%nat /\ core = INR n * pow l n.

```

```

Definition coefficient_mass_factorization (n : nat) (core q : R) : Prop :=
  (0 < n)%nat /\ core = INR (2 ^ (n - 1)) * pow q n.

```

```

Definition residual_scale_equality (n : nat) (l q : R) : Prop :=
  pow l n = pow q n.

```

```

Definition coefficient_mass_equality (n : nat) : Prop :=
  n = (2 ^ (n - 1))%nat.

```

```

Lemma real_core_factorization_unique_value

```

```

  (n : nat) (core l1 l2 : R) :
  real_core_factorization n core l1 ->
  real_core_factorization n core l2 ->
  INR n * pow l1 n = INR n * pow l2 n.

```

Proof.

```
intros [_ H1] [_ H2].
```

```
rewrite <- H1, <- H2.
```

```
reflexivity.
```

Qed.

```

Lemma real_core_factorization_substitutes_y_power

```

```

  (n : nat) (core l y : R) :
  real_core_factorization n core l ->
  pow y n = 2 * core ->
  pow y n = 2 * INR n * pow l n.

```

Proof.

```
intros [_ Hcore] Hy.
```

```
rewrite Hy, Hcore.
```

```
ring.
```

Qed.

Lemma coefficient_mass_factorization_substitutes_difference

```
(n : nat) (core q diff : R) :
coefficient_mass_factorization n core q ->
diff = 2 * core ->
diff = INR (2 ^ n) * pow q n.
```

Proof.

```
intros [Hnpos Hcore] Hdiff.
rewrite Hdiff, Hcore.
replace (INR (2 ^ n)) with (2 * INR (2 ^ (n - 1))).
- set (u := pow q n). ring.
- destruct n as [|n].
+ lia.
+ rewrite Nat.pow_succ_r' by lia.
  replace (S n - 1)%nat with n by lia.
  rewrite mult_INR.
  simpl.
  lra.
```

Qed.

Lemma two_scale_factorizations_relation

```
(n : nat) (core l q : R) :
real_core_factorization n core l ->
coefficient_mass_factorization n core q ->
pow l n = pow q n * (INR (2 ^ (n - 1)) / INR n).
```

Proof.

```
intros [Hnpos Hl] [_ Hq].
assert (Heq : INR n * pow l n = INR (2 ^ (n - 1)) * pow q n).
{ rewrite Hl in Hq. exact Hq. }
assert (Hn_pos : 0 < INR n) by (apply lt_0_INR; exact Hnpos).
apply Rmult_eq_reg_l with (r := INR n).
- rewrite Heq. field. lra.
- lra.
```

Qed.

Lemma Rpow_pos_of_pos (a : R) (n : nat) :

```
0 < a -> 0 < pow a n.
```

Proof.

```
intro Ha.
induction n as [|n IH]; simpl.
- lra.
- apply Rmult_lt_0_compat; assumption.
```

Qed.

Lemma residual_scale_equality_forces_coefficient_mass_equality

```
(n : nat) (core l q : R) :
real_core_factorization n core l ->
coefficient_mass_factorization n core q ->
residual_scale_equality n l q ->
0 < q ->
coefficient_mass_equality n.
```

Proof.

```
intros [_ Hl] [_ Hq] Hscale Hqpos.
unfold coefficient_mass_equality.
assert (Heq : INR n * pow q n = INR (2 ^ (n - 1)) * pow q n).
{ pose proof Hl as Hleft.
  unfold residual_scale_equality in Hscale.
```

```

    rewrite Hscale in Hleft.
    rewrite Hleft in Hq.
    exact Hq. }
assert (Hqpow_pos : 0 < pow q n) by now apply Rpow_pos_of_pos.
assert (Hcancel : INR n = INR (2 ^ (n - 1))).
{ apply Rmult_eq_reg_r with (r := pow q n).
  - exact Heq.
  - lra. }
apply INR_eq.
exact Hcancel.
Qed.

```

```

Lemma coefficient_mass_equality_forces_residual_scale_equality
  (n : nat) (core l q : R) :
  real_core_factorization n core l ->
  coefficient_mass_factorization n core q ->
  coefficient_mass_equality n ->
  residual_scale_equality n l q.

```

```

Proof.
  intros [Hnpos Hl] [_ Hq] Hcoeff.
  unfold residual_scale_equality.
  unfold coefficient_mass_equality in Hcoeff.
  assert (Heq : INR n * pow l n = INR n * pow q n).
  { replace (2 ^ (n - 1))%nat with n in Hq by lia.
    rewrite Hl in Hq.
    exact Hq. }
  assert (Hn_pos : 0 < INR n) by (apply lt_0_INR; exact Hnpos).
  apply Rmult_eq_reg_l with (r := INR n).
  - exact Heq.
  - lra.
Qed.

```

```

Theorem coefficient_symmetry_compatibility_iff
  (n : nat) (core l q : R) :
  real_core_factorization n core l ->
  coefficient_mass_factorization n core q ->
  0 < q ->
  (residual_scale_equality n l q <-> coefficient_mass_equality n).

```

```

Proof.
  intros Hlin Hcoef Hqpos.
  split.
  - intro Hscale.
    eapply residual_scale_equality_forces_coefficient_mass_equality;
eauto.
  - intro Hcoeff.
    eapply coefficient_mass_equality_forces_residual_scale_equality;
eauto.
Qed.

```

```

Record CoefficientSymmetryData : Type := {
  cs_n : nat;
  cs_core : R;
  cs_l : R;
  cs_q : R;
  cs_l_pos : 0 < cs_l;
  cs_q_pos : 0 < cs_q;
  cs_linear_factorization : real_core_factorization cs_n cs_core cs_l;

```

```

    cs_coefficient_factorization : coefficient_mass_factorization cs_n
cs_core cs_q;
    cs_residual_scale_equality : residual_scale_equality cs_n cs_l cs_q
}.

```

```

Lemma coefficient_symmetry_data_forces_shift (s : CoefficientSymmetryData)
:

```

```

    coefficient_mass_equality (cs_n s).

```

```

Proof.

```

```

    eapply residual_scale_equality_forces_coefficient_mass_equality.

```

```

    - exact (cs_linear_factorization s).

```

```

    - exact (cs_coefficient_factorization s).

```

```

    - exact (cs_residual_scale_equality s).

```

```

    - exact (cs_q_pos s).

```

```

Qed.

```

```

Close Scope R_scope.

```

```

(* ----- Модульная оговорка: сравнение слабее равенства ----- *)
Section ModularRemark.

```

```

Local Open Scope Z_scope.

```

```

Definition modular_congruent (modq a b : Z) : Prop :=
    exists t : Z, a = b + modq * t.

```

```

Lemma integer_equality_implies_modular_power_congruence

```

```

    (n : nat) (x y z modq : Z) :

```

```

    Z.pow x (Z.of_nat n) + Z.pow y (Z.of_nat n) =

```

```

    Z.pow z (Z.of_nat n) ->

```

```

    modular_congruent modq

```

```

    (Z.pow z (Z.of_nat n))

```

```

    (Z.pow x (Z.of_nat n) + Z.pow y (Z.of_nat n)).

```

```

Proof.

```

```

    intro H.

```

```

    exists 0%Z.

```

```

    rewrite <- H.

```

```

    ring.

```

```

Qed.

```

```

Lemma modular_congruence_has_integer_witness

```

```

    (modq lhs rhs : Z) :

```

```

    modular_congruent modq lhs rhs -> exists t : Z, lhs = rhs + modq * t.

```

```

Proof.

```

```

    exact (fun H => H).

```

```

Qed.

```

```

Lemma integer_equality_is_zero_modular_witness

```

```

    (n : nat) (x y z modq : Z) :

```

```

    Z.pow x (Z.of_nat n) + Z.pow y (Z.of_nat n) =

```

```

    Z.pow z (Z.of_nat n) ->

```

```

    exists t : Z,

```

```

    t = 0%Z /\

```

```

    Z.pow z (Z.of_nat n) =

```

```

    Z.pow x (Z.of_nat n) + Z.pow y (Z.of_nat n) + modq * t.

```

```

Proof.

```

```

    intro H.

```

```

exists 0%Z.
split; [reflexivity|].
rewrite <- H.
ring.
Qed.

```

```

Lemma modular_congruence_not_integer_equality :
  modular_congruent 5 (Z.pow 3 3) (Z.pow 1 3 + Z.pow 1 3) /\
  Z.pow 1 3 + Z.pow 1 3 <> Z.pow 3 3.

```

```

Proof.
  split.
  - exists 5%Z.
    reflexivity.
  - discriminate.
Qed.

```

```

Record ModularResidues : Type := {
  mpr_modq : Z;
  mpr_n : nat;
  mpr_m : Z;
  mpr_p : Z;
  mpr_A : Z;
  mpr_B : Z;
  mpr_z : Z;
  mpr_x : Z;
  mpr_mn_residue :
    modular_congruent mpr_modq (Z.pow mpr_m (Z.of_nat mpr_n)) mpr_A;
  mpr_pn_residue :
    modular_congruent mpr_modq (Z.pow mpr_p (Z.of_nat mpr_n)) mpr_B;
  mpr_z_residue : modular_congruent mpr_modq mpr_z (mpr_A + mpr_B);
  mpr_x_residue : modular_congruent mpr_modq mpr_x (mpr_A - mpr_B)
}.

```

End ModularRemark.

(* ----- Элементарные факты о росте на натуральных числах ----- *)
Local Open Scope nat_scope.

```

Lemma pow2_gt_linear_shift (k : nat) :
  2 ^ (k + 3) > 2 * (k + 3).

```

```

Proof.
  induction k as [|k IH]; simpl.
  - lia.
  - replace (S k + 3) with (k + 4) by lia.
    replace (2 ^ (S k + 3)) with (2 * 2 ^ (k + 3)) by
      (replace (S k + 3) with (S (k + 3)) by lia;
       rewrite Nat.pow_succ_r by lia;
       rewrite Nat.mul_comm; reflexivity).
    assert (IH' : 2 ^ (k + 3) > 2 * (k + 3)) by exact IH.
    assert (Htmp : 2 * 2 ^ (k + 3) > 2 * (2 * (k + 3))).
    { apply Nat.mul_lt_mono_pos_l; [lia|exact IH']. }
    replace (2 * (2 * (k + 3))) with (4 * (k + 3)) in Htmp by lia.
    apply Nat.le_lt_trans with (m := 4 * (k + 3)).
    + lia.
    + exact Htmp.
Qed.

```


Lemma pow2_gt_linear (n : nat) :

$3 \leq n \rightarrow 2^n > 2 * n$.

Proof.

intros Hn.

destruct (Nat.le_exists_sub 3 n Hn) as [k [Hk _]].

rewrite Hk.

replace (3 + k) with (k + 3) by lia.

apply pow2_gt_linear_shift.

Qed.

Lemma pow_eq_linear_cases (n : nat) :

$2^n = 2 * n \rightarrow n = 0 \vee n = 1 \vee n = 2$.

Proof.

destruct n as [|n].

- simpl. intro H. now left.

- destruct n as [|n].

+ simpl. intro H. right; left; lia.

+ destruct n as [|n].

* simpl. intro H. right; right; lia.

* intro H.

assert (3 ≤ S (S (S n))) by lia.

specialize (pow2_gt_linear _ H0) as Hgt.

rewrite H in Hgt. lia.

Qed.

Lemma pow_eq_linear_positive (n : nat) :

$2^n = 2 * n \rightarrow n = 1 \vee n = 2$.

Proof.

intro H.

destruct (pow_eq_linear_cases n H) as [H0 | [H1 | H2]].

- subst n. discriminate.

- now left.

- now right.

Qed.

Lemma two_pow_eq_two_mul_iff_shift (n : nat) :

$(2^n = 2 * n) \% \text{nat} \leftrightarrow (n = 2^{(n-1)}) \% \text{nat}$.

Proof.

destruct n as [|n].

- simpl. split; intro H; lia.

- split.

+ intro H.

rewrite Nat.pow_succ_r' in H by lia.

apply Nat.mul_cancel_l in H; [|lia].

replace (S n - 1) % nat with n by lia.

symmetry.

exact H.

+ intro H.

rewrite Nat.pow_succ_r' by lia.

replace (S n - 1) % nat with n in H by lia.

rewrite H.

reflexivity.

Qed.

Lemma binary_scaling_roots_only_one_two (n : nat) :

$(n = 2^{(n-1)}) \% \text{nat} \rightarrow n = 1 \% \text{nat} \vee n = 2 \% \text{nat}$.

Proof.

```

    intro Hshift.
    apply two_pow_eq_two_mul_iff_shift in Hshift.
    now apply pow_eq_linear_positive in Hshift.
Qed.

Lemma coefficient_symmetry_data_forces_small_exponent
  (s : CoefficientSymmetryData) :
  cs_n s = 1%nat \ / cs_n s = 2%nat.
Proof.
  apply binary_scaling_roots_only_one_two.
  apply coefficient_symmetry_data_forces_shift.
Qed.

Theorem coefficient_symmetry_data_excludes_high_exponent
  (s : CoefficientSymmetryData) :
  (2 < cs_n s)%nat -> False.
Proof.
  intro Hgt.
  destruct (coefficient_symmetry_data_forces_small_exponent s) as [H1|H2];
lia.
Qed.

Theorem coefficient_symmetry_compatibility_excludes_high_exponent
  (n : nat) (core l q : R) :
  real_core_factorization n core l ->
  coefficient_mass_factorization n core q ->
  residual_scale_equality n l q ->
  (0 < q)%R ->
  (2 < n)%nat -> False.
Proof.
  intros Hlin Hcoef Hscale Hqpos Hgt.
  assert (Hcoeff : coefficient_mass_equality n).
  { eapply residual_scale_equality_forces_coefficient_mass_equality;
eauto. }
  unfold coefficient_mass_equality in Hcoeff.
  destruct (binary_scaling_roots_only_one_two n Hcoeff) as [H1|H2]; lia.
Qed.

Goal forall s : CoefficientSymmetryData, cs_n s = 1%nat \ / cs_n s = 2%nat.
Proof.
  exact coefficient_symmetry_data_forces_small_exponent.
Qed.

```

Время вычисления: 2,7 с

FLT on □ main [?] ...

→ coqc GlobalNormalization.v

FLT on □ main [?] took 2.7s ...